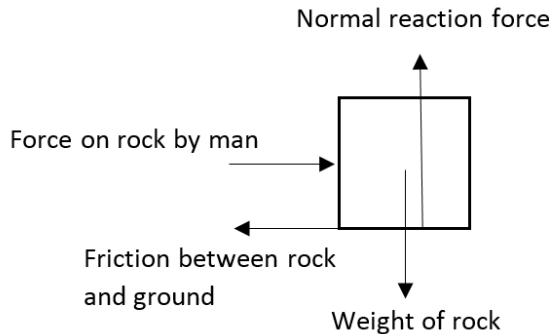


2(a)(i) The student's argument is not correct because the reaction force acts on the man, i.e. the reaction force on the man and the force the man exerts on the rock are acting on two different bodies, so they do not cancel out.

(ii)



Note: The normal reaction force and weight of rock must form an anticlockwise couple, to balance the clockwise couple from the horizontal forces.

2(b)(i) Block A and B moves with the same velocity when maximum compression of the spring takes place.

(ii) By the conservation of momentum,
total initial momentum = total final momentum

$$m_A u_A + m_B v_B = (m_A + m_B) v$$

$$(2.0)(10) + (3.0)(5.0) = (2.0 + 3.0)v$$

$$v = 7.0 \text{ m s}^{-1}$$

(iii) Let maximum compression be x .

By conservation of energy,

loss in KE of block A = gain in EPE in the spring + gain in KE of block B

$$\frac{1}{2} m_A (u_A^2 - v^2) = \frac{1}{2} kx^2 + \frac{1}{2} m_B (v^2 - u_B^2)$$

$$\frac{1}{2} (2.0)(10^2 - 7.0^2) = \frac{1}{2} (500)x^2 + \frac{1}{2} (3.0)(7.0^2 - 5.0^2)$$

$$x = 0.24 \text{ m}$$